```
clear; clc; close all;

Fs = 8000;
fc = 1500;
wc = fc / (Fs/2);
N = 50;

% Window Functions
w_rect = rectwin(N+1);
w_hamm = hamming(N+1);
w_hann = hann(N+1);
w_black = blackman(N+1);

% FIR Filter Design
b_rect = fir1(N, wc, w_rect);
b_hamm = fir1(N, wc, w_hamm);
b_hann = fir1(N, wc, w_hann);
b_black = fir1(N, wc, w_black);

% Frequency Response
[H_rect, f] = freqz(b_rect, 1, 1024, Fs);
[H_hamm, ~] = freqz(b_hamm, 1, 1024, Fs);
[H_hann, ~] = freqz(b_hann, 1, 1024, Fs);
[H_black, ~] = freqz(b_black, 1, 1024, Fs);

% Display Numerical Outputs
fprintf('\n=== EXPERIMENT 4: FIR WINDOWING TECHNIQUES ===\n\n');
```

```
=== EXPERIMENT 4: FIR WINDOWING TECHNIQUES ===
```

```
fprintf('Filter Order (N): %d\n', N);
```

```
Filter Order (N): 50
```

```
fprintf('Cutoff Frequency: %d Hz\n\n', fc);
```

```
Cutoff Frequency: 1500 Hz
```

```
fprintf('--- RECTANGULAR WINDOW COEFFICIENTS (First 5) ---\n');
```

```
--- RECTANGULAR WINDOW COEFFICIENTS (First 5) ---
```

```
disp(b_rect(1:5));
```

```
   -0.0118         0    0.0128    0.0103   -0.0058
```

```
fprintf('--- HAMMING WINDOW COEFFICIENTS (First 5) ---\n');
```

```
--- HAMMING WINDOW COEFFICIENTS (First 5) ---
```

```
disp(b_hamm(1:5));
```

```
   -0.0009         0    0.0012    0.0011   -0.0008
```

```
fprintf('--- HANNING WINDOW COEFFICIENTS (First 5) ---\n');
```

```
--- HANNING WINDOW COEFFICIENTS (First 5) ---
```

```
disp(b_hann(1:5));
```

```
   1.0e-03 *

         0         0    0.2009    0.3592   -0.3588
```

```
fprintf('--- BLACKMAN WINDOW COEFFICIENTS (First 5) ---\n');
```

```
--- BLACKMAN WINDOW COEFFICIENTS (First 5) ---
```

```
disp(b_black(1:5));
```

```
   1.0e-03 *

         0         0    0.0743    0.1374   -0.1434
```

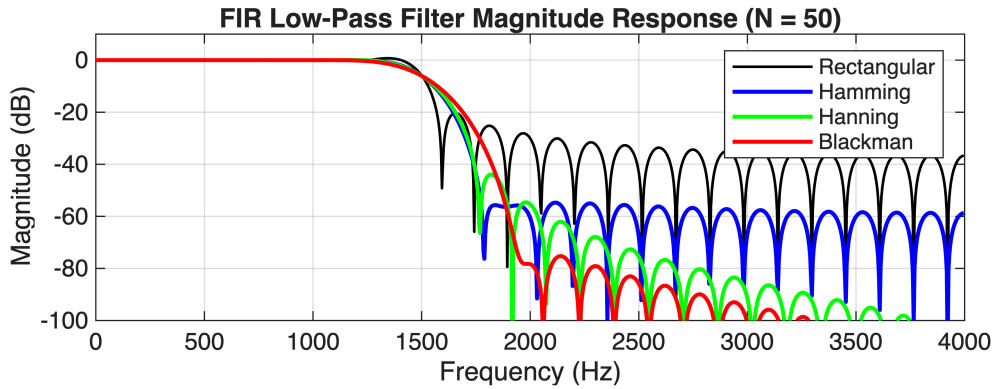
```
% Plotting
figure('Name', 'FIR Windowing Techniques Comparison');

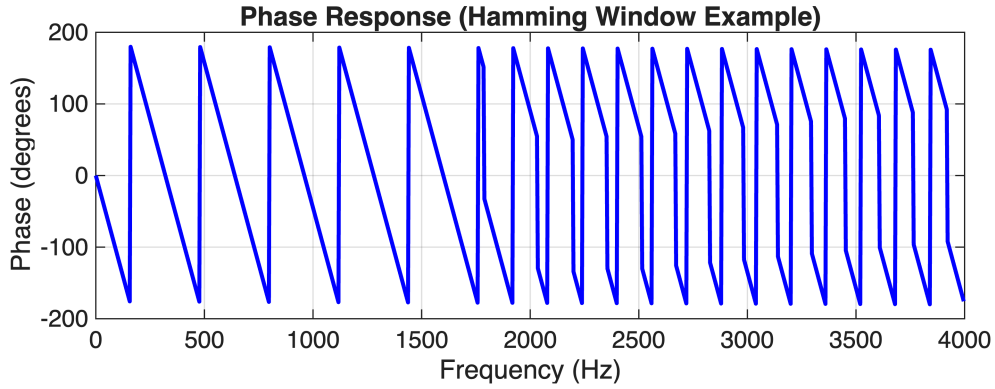
subplot(2,1,1);

plot(f, 20*log10(max(abs(H_rect),1e-10)), 'k', 'LineWidth', 1);
hold on;
plot(f, 20*log10(max(abs(H_hamm),1e-10)), 'b', 'LineWidth', 1.5);
plot(f, 20*log10(max(abs(H_hann),1e-10)), 'g', 'LineWidth', 1.5);
plot(f, 20*log10(max(abs(H_black),1e-10)), 'r', 'LineWidth', 1.5);

grid on;
legend('Rectangular', 'Hamming', 'Hanning', 'Blackman');
title('FIR Low-Pass Filter Magnitude Response (N = 50)');
xlabel('Frequency (Hz)');
ylabel('Magnitude (dB)');
ylim([-100 10]);

subplot(2,1,2);
plot(f, angle(H_hamm)*180/pi, 'b', 'LineWidth', 1.5);
grid on;
title('Phase Response (Hamming Window Example)');
xlabel('Frequency (Hz)');
ylabel('Phase (degrees)');
```


FIR Low-Pass Filter Magnitude Response (N = 50)
Magnitude (dB)
Frequency (Hz)
Rectangular
Hamming
Hanning
Blackman
0
-20
-40
-60
-80
-100
0
500
1000
1500
2000
2500
3000
3500
4000


Phase Response (Hamming Window Example)
Phase (degrees)
Frequency (Hz)
200
100
0
-100
-200
0
500
1000
1500
2000
2500
3000
3500
4000